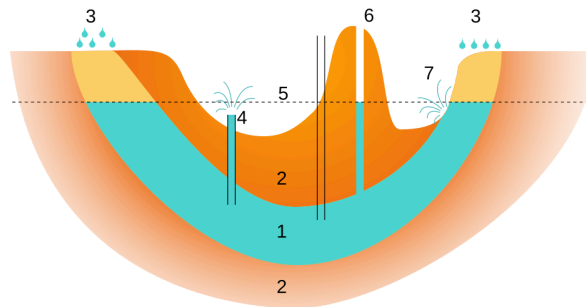


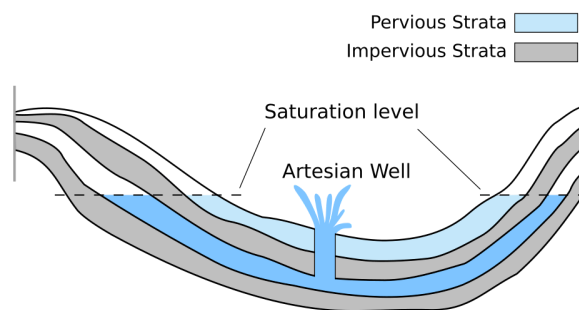
Artesian well

An **artesian well** is a **well** that brings **groundwater** to the surface without pumping because it is under pressure within a body of rock or sediment known as an **aquifer**.^[1] When trapped water in an aquifer is surrounded by layers of **impermeable** rock or clay, which apply positive pressure to the water, it is known as an **artesian aquifer**.^[1] If a well were to be sunk into an artesian aquifer, water in the well-pipe would rise to a height corresponding to the point where **hydrostatic equilibrium** is reached.^[1]

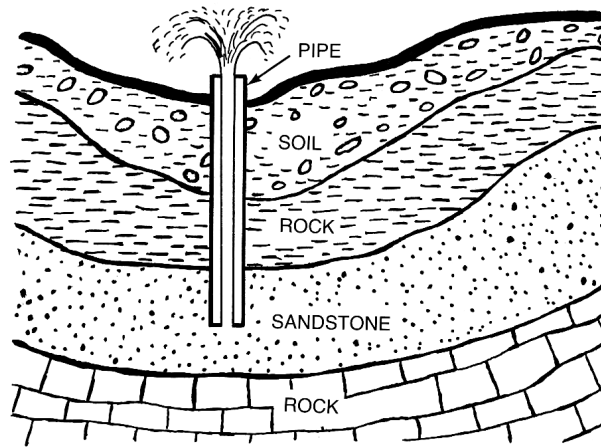


Artesian aquifer scheme:

1. **Aquifer**
2. Impervious strata
3. Infiltration area
4. Artesian well
5. Saturation level
6. Subartesian well
7. Artesian spring



Geological **strata** giving rise to an artesian well



Schematic of an artesian well



U.S. Navy [Seabees](#) tapping an artesian well in [Helmand Province](#), Afghanistan

A well drilled into such an aquifer is called an *artesian well*.^[1] If water reaches the ground surface under the natural pressure of the aquifer, the well is termed a *flowing artesian well*.^{[2][3]}

[Fossil water](#) aquifers can also be artesian if they are under sufficient pressure from the surrounding rocks, similar to how many newly tapped [oil wells](#) are pressurized.



Artesian well near the Myllylähde spring in [Alastaro](#), Finland

Not all aquifers are artesian (i.e., [water table](#) aquifers occur where the groundwater level at the top of the aquifer is at equilibrium with [atmospheric pressure](#)). Aquifers recharge when the water table at its recharge zone is at a higher [elevation](#) than the head of the well.

History

The first mechanically accurate explanation for artesian wells was given by [Al-Biruni](#).^[4] Artesian wells were named after [Artois](#) in [France](#), where many artesian wells were drilled by [Carthusian](#) monks from 1126.^[5]

See also

- [Fluid mechanics](#)
- [Great Artesian Basin](#)
- [Hydrogeology](#)
- [Qanat](#)
- [Spring \(hydrology\)](#)
- [List of artesian wells in the United States](#)
- [Drilled wells](#) (for bore water mechanics)

References

1. "Artesian Water and Artesian Wells" (https://www.usgs.gov/special-topic/water-science-school/science/artesian-water-and-artesian-wells?qt-science_center_objects=0#qt-science_center_objects) . *United States Geological Survey*. 2018-06-06. Retrieved 2020-12-18.
2. Wheeler, H. W. (1980), *Artesian bores of South Australia: an annotated photographic record, 1939–1948*, Pioneer Books, [ISBN 978-0-908065-06-6](#)
3. Federal Water Resources Assistance Program (Australia); New South Wales Department of Water Resources, Technical Services Division; Australian Water Resources Council. Interstate Working Group on the Great Artesian Basin (1990), *Specification for construction, reconditioning or plugging of bores tapping recognised aquifers of the Great Artesian Basin in New South Wales* (<http://trove.nla.gov.au/work/6766983>) (1st ed.), Dept. of Water Resources, Technical Services Division, retrieved 19 January 2014

4. Davis, Stanley N.; De Wiest, Roger J. M. (January 15, 1966). *Hydrogeology*. Wiley. pp. 9–10. ISBN 9780471199007.
5. Frances Gies and Joseph Gies, *Cathedral, Forge, and Waterwheel* subtitled "Technology and Invention in the Middle Ages". Harper Perennial, 1995. ISBN 0-06-016590-1, page 112.

External links

- [Video of an artesian aquifer releasing water under its own pressure \(http://www.daleswater.co.uk/what-is-an-artesian-well/\)](http://www.daleswater.co.uk/what-is-an-artesian-well/)